

Andrew J. Brown

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Skills

Programming: Python, Rust, R, SQL, TypeScript, Julia, Matlab, C++/C

Data Processing & Visualization: Polars, Pandas, Apache Arrow, Holoviews/Panel Ecosystem, R/Python Shiny, R Tidyverse, Xarray/Zarr, Pyvista/VTK, Matplotlib/Seaborn, ggplot2

Scientific Computing & Modeling: Scikit-Learn, Darts, Scipy, statsmodels, CUDA, Numba, Dask

Work Experience

Water Systems Data Analyst

Milwaukee Metropolitan Sewerage District; Milwaukee, WI; September 2024 — Present

- Developing and managing data systems collecting 10 billion raw records per year.
- Architecting, developing, deploying, and maintaining infrastructure-critical code.
- Developing a bespoke analysis API for the organization.
- Creating dashboards with Shiny and Panel.
- Modernizing and simplifying the organization's technological architecture.
- Built data storage and processing tools for high density remote sensing data products (20 billion observations per day).
- Designing APIs for Python and R using Rust.
- Developing, deploying, and maintaining automated QA/QC procedures for tabular time series data.
- Creating Statistical and physics-based models to predict behavior and outcomes in a complex, networked system.
- Parallel and concurrent software development.

Senior Water Resources Management Specialist

Wisconsin Department of Natural Resources; Madison, Wisconsin; September 2023 — September 2024

- Performed scientific analysis for a highly controversial (34,000 public comments), complex linear project (41 miles, 1,000+ wetland and stream crossings).
- Developed, maintained, and queried a cross-disciplinary (5 datasets, 200,000,000 rows) database (Python, SQLite).
- Piloted machine-learning models (MaxEnt, GLM, random forest, SVM) and processing pipelines (Polars, Dask, Xarray) for fish species distribution modeling (Numpy, Numba, Scipy, Scikit-Learn).
- Created automated processing tools to transform large, complex datasets into clean, communicative visualizations and data products.
- Delivered key analysis and writing (100 pages, 35,000 words) for a high-scrutiny context area while working remotely.

Education

M.S. Environmental Science - Conc. Water Resources

Indiana University, Bloomington

08/21/2021 - 05/08/2023

GPA: 3.957

B.S. Geography

Indiana University, Bloomington

08/21/2017 - 05/08/2021

GPA: 3.886